

Kartikey Bhatnagar

Portfolio: kartikeycodes.tech

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Technical Skills

Languages: TypeScript, JavaScript, Python, SQL

Frontend: Next.js, React, TanStack Query, Zustand

Backend: Node.js, Express, tRPC, Inngest, Zod, WebSockets, Prisma

AI & Vector: LangChain, pgvector, Tree-sitter, Gemini API, AssemblyAI

Databases: PostgreSQL, MongoDB, Redis

Infrastructure: Docker, RabbitMQ, AWS, GitHub Actions

Projects

YSync | *TypeScript, Node.js, WebSockets, React, PostgreSQL, Redis, Prisma, IndexedDB* **June 2026**

- **Implemented custom sequence CRDT** for real-time collaborative text editing (RGA variant via Timestamped Insertion List), enabling conflict-free concurrent edits across replicas, verified with property-based tests simulating **5,000+** out-of-order operations
- **Built WebSocket sync server** with per-document **awareness/presence broadcasting** (cursors, selections), fanning out ops to **50+** concurrent clients with sub-**20ms** propagation latency
- **Engineered offline-first sync** merging local edit queues on reconnect via CRDT causal ordering, achieving **zero data loss** across **250** conflicting offline edit scenarios
- **Persisted CRDT operation log to PostgreSQL** with periodic state snapshots and tombstone garbage collection, bounding storage growth to **85%** of naïve append-only storage

Kyron | *Next.js, Inngest, Tree-sitter, pgvector, Gemini SDK, PostgreSQL, Octokit* **March 2026**

- **Engineered durable PR review pipeline** via **Inngest**, processing diffs through **Tree-sitter AST chunking**, **768-dim pgvector** retrieval, and **Gemini**-based review, posting comments in **45s** for PRs **under 500 lines**, with each stage **checkpointed** for crash recovery
- **Built incremental indexing** via **GitHub Compare API** to re-embed only modified chunks, cutting embedding API calls by **~80%** on typical commits, while **HNSW indexing** sustains **sub-20ms** similarity search as chunk count scales
- **Absorbed Gemini rate limits** using **exponential backoff with jitter** and **parallel batching of 15 chunks**, reducing total embedding time by **~4x** vs. sequential calls while sustaining throughput without quota exhaustion
- **Eliminated race condition** in concurrent PR indexing via composite unique constraints and upsert writes on (**repo_id, commit_sha, chunk_hash**), verified with concurrent-load test across **64** simultaneous jobs, preventing duplicate embeddings at scale

SQLFlow | *Next.js, TypeScript, Express.js, PostgreSQL, MongoDB, Monaco Editor* **Jan 2026**

- **Engineered a multitenant SQL sandbox execution engine** provisioning isolated per-user PostgreSQL schemas on demand, dynamically generating DDL/DML from JSON-defined assignment schemas to seed sample data and execute untrusted student queries under search-path scoping and keyword-based validation
- **Built an order-independent, type-normalizing result-set comparator** for grading SQL submissions, correctly handling row ordering, duplicate rows, and cross-type value coercion against expected output avoiding brittle query-text or string matching
- **Architected a dual-database system** separating MongoDB (assignments, per-user progress, execution/hint history) from PostgreSQL (isolated sandbox execution), with **TTL-indexed** audit logs (7-day execution, 30-day hint history) bounding unbounded log growth

Experience

Open Source Contributor | **ElixirTechCommunity/ElixirV3**

- **Reduced page load time by 25%** via **lazy loading**, **unused CSS elimination**, and **render-blocking script deferral**; measured via **Lighthouse CI** on PR, not one-off local scores.
- Resolved **LCP regression** with **priority hints** on hero images; eliminated **CLS** from implicit image dimensions; deferred **third-party scripts** to reduce **TBT** on initial load.

Education

ABESEC

Electronics and Communication Engineering

Aug. 2023 – Aug. 2027

Expected Aug. 2027 | Ghaziabad, India